Phase 8: Project Demonstration

1. Live Deployment

The application has been successfully deployed to the cloud. Because the repository structure was designed to maintain root-level execution files (`Procfile`, `requirements.txt`), the live link functions exactly as expected without requiring redeployment.

(Insert deployed URL here)

2. Demonstration Scenarios

Scenario 1: High-Risk Alert (Early Warning)

- **Inputs:** Extremely high Annual Rainfall (e.g., 3000mm), High Humidity (90%), High Cloud Cover.
- **System Action:** The XGBoost model processes the data array.
- **Output:** The UI flashes a red warning indicating **"Flood Likely"** with a probability of >85%.

Scenario 2: Normal Conditions

- **Inputs:** Moderate Rainfall (e.g., 500mm), Average Humidity (50%), Low Cloud Cover.
- **System Action:** The XGBoost model processes the data array.
- **Output:** The UI displays a green banner indicating **"No Flood Risk"** with a probability of <20%.

Scenario 3: Meteorologist Validation

- A meteorologist inputs historical data from a known flood event in 2015.
- The system accurately predicts the flood, validating the model's 96.55% accuracy metric.

3. Future Enhancements for Demonstration

- Integration with live weather APIs (e.g., OpenWeatherMap) to auto-fill the form based on GPS location.
- Interactive maps showing regional risk zones.